

RESEARCH PAPER

MAUVE: Cold neutral gas in the outflow of NGC 4383 and evidence for a fountain flow

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Abstract

We present a multiphase study of the star-formation-driven outflow in the Virgo galaxy NGC 4383, combining new ALMA CO(2–1) data with deep MeerKAT H_I imaging and MUSE spectroscopy obtained as part of the Multiphase Astrophysics to Unveil the Virgo Environment (MAUVE) program. Our previous work revealed a spectacular ionised outflow, but the role of the cold phase remained unclear. Our analysis shows that molecular gas is affected only within the inner ~ 1 kpc above the disc, where CO clouds exhibit disturbed kinematics and spatial correspondence with the ionisation cone. Beyond this, the CO surface brightness rapidly drops, indicating that the molecular phase contributes little at larger heights. In contrast, the H_I distribution shows plumes aligned with the ionised cone and complex kinematics suggestive of parts of the atomic phase being entrained in the outflow. However, the extended and warped H_I disc associated with NGC 4383 complicates the unambiguous identification of outflowing atomic gas and, most importantly, the quantification of outflowing mass and loading factor. Independent support for a cold component in the outflow comes from dust extinction features coincident with H_I plumes a few kiloparsecs above the disc. Together, despite significant uncertainties, these results suggest that the atomic phase likely dominates the cold outflow above ~ 1 kpc. The observed cold gas velocities remain well below the velocities of the ionised phase, favouring a scenario in which NGC 4383 hosts not a large-scale escaping wind but a galactic fountain in which feedback simply redistributes material within the halo, regulating both current and future star formation.

1. Introduction

How gas cycles in and out of galaxies is key for understanding their life cycle. Indeed, it is the continuous circulation of baryons in the halo that truly regulates the star formation history of galaxies (Tumlinson et al., 2017; Saintonge & Catinella, 2022).

Historically, this cycle was seen as composed of two relatively distinct processes: inflows were thought to be associated with gas accretion from satellites or cold filaments (e.g., Kereš et al., 2005; Sancisi et al., 2008), while outflows were linked to feedback from supernovae or accreting supermassive black holes (e.g., Hopkins et al., 2012). In this classical picture, inflows and outflows can be seen as parts of the baryon cycle driven and regulated by very different physical mechanisms.

However, growing evidence in the local Universe suggests that this dichotomy is overly simplistic, at least at current epochs. On the one hand, feedback-driven winds fail to reach the escape velocity of their host galaxies, especially when driven by star formation, making it unlikely that they permanently eject large masses of gas into the intergalactic medium (e.g., Leroy et al., 2015; Concas et al., 2017; McQuinn et al., 2019; Roberts-Borsani & Saintonge, 2019; Marasco et al., 2023; Mazzilli Ciraulo et al., 2025). On the other hand, direct evidence for cold filamentary inflows has never been obtained in the local Universe, and the accretion of satellites does not seem to be able account for the sustained star formation observed in most galaxies at late times (e.g., Sancisi et al., 2008; Di Teodoro & Fraternali, 2014). Instead, a galactic fountain

model, in which gas cycles between the disc and halo before returning to fuel future star formation, is gaining significant traction (Fraternali & Binney, 2008; Marinacci et al., 2010). In this view, outflows and inflows are not separate, but different phases of the same circulation process.

Probing this cycle observationally requires tracing the different phases of the interstellar medium (ISM) as they leave and re-enter galactic discs. While the warm gas component is relatively straightforward to map thanks to recombination lines in the rest-frame optical spectrum, this phase is not expected to dominate the mass budget of outflows (e.g., Veilleux et al., 2020 and references therein). Instead, most of the mass is expected to reside in colder phases, either molecular or atomic (e.g., Bolatto et al., 2013b; Cicone et al., 2014; Leroy et al., 2015; Fluetsch et al., 2019; Avery et al., 2022). Molecular gas traced by CO has been observed in a number of nearby normal star-forming systems, but typically only within the inner kiloparsec (Bolatto et al., 2013b; Leroy et al., 2015; Noon et al., 2023), consistent with expectations from numerical models (e.g., Vijayan & Krumholz, 2024). Atomic hydrogen, by contrast, is expected to be the dominant component on kiloparsec scales, but direct evidence for its participation in galactic outflows is still very limited (e.g., McClure-Griffiths et al., 2018; Martini et al., 2018; Heyer et al., 2025). This is due to a combination of sensitivity and resolution challenges, as well as the intrinsic complexity of H_I morphology, which often includes thick discs and warps that complicate the identification of extra-

planar material (Marasco et al., 2019) and might systematically overestimate any quantification of the mass of gas in process of leaving (or re-entering) the star-forming disc.

Therefore, progress in this field requires a substantial increase in the number of galaxies for which multiphase outflows can be studied at sufficient spatial resolution to disentangle disc and extraplanar gas. To this end, we are conducting the MAUVE project (Multiphase Astrophysics to Unveil the Virgo Environment), which is designed to trace the baryon cycle of galaxies as they fall into the Virgo cluster. At the time of writing, MAUVE is collecting observations from VLT/MUSE, ALMA, and HST to characterise the gas cycle in unprecedented detail in 40 galaxies in the Virgo cluster, achieving spatial resolutions ranging from ~ 100 to ~ 6 pc scales, depending on the facility (Catinella et al. submitted). This approach will allow us to probe the baryon cycle and determine how it is affected by the cluster environment at different stages of infall into the cluster. Particularly key is to understand how star formation is quenched in satellites after the outer gas disc has been stripped (Cortese et al., 2021). MAUVE has already uncovered striking examples of warm ionised outflows at different stages of cluster infall, ranging from NGC 4383 (Watts et al., 2024), which has only recently begun its descent into Virgo, to the post-pericentre, H α -deficient system NGC 4064 (Attwater et al., submitted).

In this paper, we present the first MAUVE multiphase study of a star-formation-driven outflow, focusing on NGC 4383. This galaxy is the most actively star-forming galaxy in the whole MAUVE sample, as measured from its distance from the locus of the star-forming main sequence. Following Leroy et al. (2019), NGC4383 has an integrated stellar mass of $\sim 10^{9.4} M_{\odot}$ and a current star formation rate of $\sim 1 M_{\odot} \text{ yr}^{-1}$, positioning it ~ 0.5 dex above the main sequence. Our MAUVE-MUSE observations revealed a powerful ionised outflow extending several kiloparsecs above the disc (Watts et al., 2024), while earlier single-dish and interferometric studies established the presence of an extended and warped H α disc (Chung et al., 2009). Here, we combine new ALMA CO(2–1) observations with MeerKAT H α imaging and MUSE spectroscopy to jointly investigate the molecular, atomic, and ionised gas components of the outflow. Our aim is to evaluate the role of the cold phases in the baryon cycle of NGC 4383 and to test whether its outflow is better described as a classical galactic wind or as part of a fountain-like circulation.

This paper is organised as follows. Section 2 describes the observations and data reduction. Sections 3 present the analysis of the molecular and atomic gas properties of NGC 4383, with direct comparison to the ionised component. Section 4 discusses the implications for gas cycling and feedback in NGC 4383, and Section 5 summarises our conclusions. Throughout this work we adopt a distance to NGC 4383 of 16.5 Mpc (Mei et al., 2007), for which 1 arcsecond corresponds to 80 pc.

2. Data

2.1 ALMA CO (2–1) data

The ALMA CO (2–1) observations for NGC 4383 were acquired as part of the MAUVE-ALMA program (2023.1.00026.S, P.I. J. Sun), which builds on the VERTICO survey (Brown et al., 2021). These data incorporate observations with both the ALMA 12m array and the Atacama Compact Array (ACA) to recover emission on all spatial scales. The observation field of view covers the star-forming inner disk of the galaxy out to 3–4 kpc along the major axis. The spectral tuning covers the full velocity range of the ^{12}CO (2–1) line emission at a native velocity resolution of 0.8 km/s.

The data reduction scheme closely follows Leroy et al. (2021) and is described in full detail in J. Sun et al. (in preparation). Briefly, we start with calibrated visibility data supplied by the observatory and verify that no obvious pathologies are present in them. We then extract spectral channels within ± 250 km/s of the systematic velocity for the ^{12}CO (2–1) line, rebin to a 2.5 km/s channel width, combine the 12m and 7m interferometry data, and image them following the PHANGS-ALMA imaging scheme (Leroy et al., 2021). For the imaging step, we adopt a Briggs weighting with a robustness parameter of 1.0 to achieve an optimal balance between sensitivity and resolution. Finally, we correct the cleaned data cube for the primary beam response pattern, convolve it to a round beam of 1 arcsec in FWHM (80 parsec at the distance of Virgo), and feather it with single-dish data to obtain the final data cube.

We further produce CO line integrated intensity (moment-0) and velocity centroid (moment-1) maps from the data cube, adopting the same set of recipes as Leroy et al. (2021). For the analyses in this work, we specifically use the ‘broad’ moment-0 maps and the moment-1 maps ‘with a prior’. These maps combine CO detection thresholds over a wide range of spatial scales and, therefore, are more inclusive to diffuse CO emission (at the expense of increased noise levels). This is important for comparisons with diffuse H α emission in the outflow from NGC 4383.

These data allow us to reach an integrated brightness of $\sim 0.9 \text{ K km s}^{-1}$, which corresponds to a molecular gas surface density of $\sim 5 M_{\odot} \text{ pc}^{-2}$ assuming a CO(2–1)-to-CO(1–0) flux ratio of 0.8 and a CO(1–0)-to-H $_2$ conversion factor typical of the Milky Way disc (i.e., $4.35 M_{\odot} \text{ pc}^{-2} \text{ K km s}^{-1}$, Bolatto et al., 2013a).

2.2 MeerKAT H α data

NGC 4383 was observed with MeerKAT in April 2023 (ID: MKT-22034, P.I. Watts). The observation was taken in L-band (856 MHz bandwidth centred at 1284 MHz) at 32K spectral resolution (26.123 kHz wide channels, corresponding to 5.6 km s^{-1} at the frequency of NGC4383).

Two tracks (rising and setting) were obtained on 03/04/23 and 17/04/23, alternating 30-min target scans with 2-min gain calibrator scans on J1120+1420. Bandpass calibration was performed at the start and end of each track using J0408–6545 and J1939–6342. Observations were scheduled at least 1.5 hours after sunset to minimise solar interference, yielding a

total observing time of 9h 12m (including overheads).

The two MeerKAT tracks were reduced using a custom version of the iDiA PROCESSMEERKAT pipeline^a (see Mad-dox et al., 2021). Each track was calibrated and continuum-subtracted separately before being combined for H_I imaging. Standard flagging and cross-calibration were applied using the observed flux, bandpass and gain calibrators. Continuum subtraction was then performed using a combination of self-calibration and polynomial fitting of line-free channels. After subtraction, the two tracks were merged and imaged with CASA using multiscale cleaning and robust weighting. A source mask was then created with SoFiA2 (Serra et al., 2015; Westmeier et al., 2021) and applied in the final deep cube reconstruction. The resulting cube has a synthesised beam of 12×7.3 arcsec² ($\sim 960 \times 584$ parsec²), a typical rms of 0.2 mJy beam⁻¹ and is binned to 2 arcsec pixel size to Nyquist-sample the beam.

We extracted moment maps (integrated intensity, flux-weighted line-of-sight velocity and effective width) for the final data cube using the Python MASKMOMENT package, developed by Tony Wong^b. We required detections to exceed 2σ in at least two consecutive channels and to cover a minimum area of twice the synthesised beam. Since this study focuses on the high-surface-density regions of the H_I disc, we note that the precise mask definition for extracting moment maps does not affect our results. We favour the effective line width (defined as the ratio of the moment zero to the peak intensity multiplied by $\sqrt{2\pi}$) as it is more robust for weak signals.

2.3 MUSE and optical imaging data

The MAUVE-MUSE observations of NGC 4383 and their data reduction are presented in Watts et al. (2024). For this work, the science-ready cube has been re-reduced using an updated version of PYMUSEPIPE (v2.28.2; Emsellem et al., 2022) as part of the MAUVE-MUSE internal data release v2 (see also Brown et al. submitted), but the differences between the two cubes are negligible (i.e. well within 1–2%).

The emission-line flux and velocity maps used here were obtained with the new Galaxy IFU Spectroscopy Tool pipeline (nGIST; Fraser-McKelvie et al., 2025b,a). nGIST builds on the GIST software developed by Bittner et al. (2019) and provides a flexible interface to the penalized pixel-fitting code PPXF (Cappellari & Emsellem, 2004; Cappellari, 2017, 2023). The emission-line maps were generated by running nGIST on data Voronoi-binned to a target S/N = 20 in the 4800–7000 Å wavelength range. This setting provides a good compromise between retaining the native resolution (1.3 arcsec, corresponding to 100 pc at 16.5 Mpc) in the inner disc of the galaxy where the high resolution is key for a proper comparison with the ALMA data, while ensuring robust continuum subtraction in the extraplanar regions. As we will see, the extra-planar region will be primarily used for the comparison with the H_I and this binning scheme guarantees that even the largest bin

at the edge of the MUSE field of view are always smaller than the synthesised beam of the MeerKAT observations.

Following the approach described by Emsellem et al. (2022), we adopted the Medium-resolution INT Library of Empirical Spectra (MILES; Vazdekis et al., 2010) to first fit the stellar kinematics while masking emission lines, and subsequently fit the emission lines while keeping the stellar kinematic parameters fixed to the values obtained in the first step.

We complement our MUSE dataset with broad-band and narrow-band imaging covering the full region mapped by our MeerKAT data. The narrow-band H α + [NII] net image was obtained as part of the Virgo Environmental Survey Tracing Ionised Gas Emission (VESTIGE; Boselli et al., 2018, 2023), while *u*, *g*, *i*, and *z* broad-band images were taken from the Next Generation Virgo Cluster Survey (NGVS; Ferrarese et al., 2012).

All velocity fields presented in this work are given in the Kinematic Local Standard of Rest (LSRK) reference frame and adopt the optical velocity definition.

3. Results

3.1 The impact of the starburst on the molecular gas disc

We begin our analysis by examining the CO(2–1) properties of NGC4383. Figure 1 shows the CO (2–1) integrated intensity contours overlaid on the NGVS optical broad-band color composite (left) and the VESTIGE H α + [NII] (right) narrow-band images. The CO emission closely follows the distribution of the dust lanes visible in optical along the major axis of the stellar disc. There are some hints suggesting that the CO disc may exhibit a warped structure similar to what observed in H_I (Chung et al., 2009 and see Sec. 3.2) towards the south-west and north east edges of the ALMA field of view. When compared to the morphology of the ionised gas outflow, the two phases display strikingly different morphologies, with the CO emission extending no more than ~ 1 kpc away from the plane of the galaxy. At face-value, no CO emission is observed in the regions classified as ‘outflow-dominated’ by Watts et al. (2024), based on the velocity dispersion of the ionised gas in NGC4383.

Despite the absence of extraplanar CO emission extending several kiloparsecs above the disk, and the lack of detailed correspondence between the 2D morphology of the CO and H α distributions, the high resolution of our CO data reveals the presence of small plumes and potentially detached clouds extending perpendicular to the disc out to distances of 0.5–1 kpc. To better assess the nature of these features, and to determine whether the molecular gas phase is at all affected by the outflow, the next step is to compare the kinematics of the ionised and molecular gas phases.

In Figure 2 we compare the CO(2–1) and MUSE H α line-of-sight velocity maps. The same colormap and velocity interval are adopted to facilitate a direct comparison between the two. To further guide the eye, we overlay the velocity contours of the ionised gas on top of the CO map, with the thick contours approximately marking where the velocity field of the ionised gas begins to exhibit the typical pattern

^a<https://github.com/spectram/pipelines/tree/HI-dev>

^b<https://github.com/tonywong94/maskmoment>

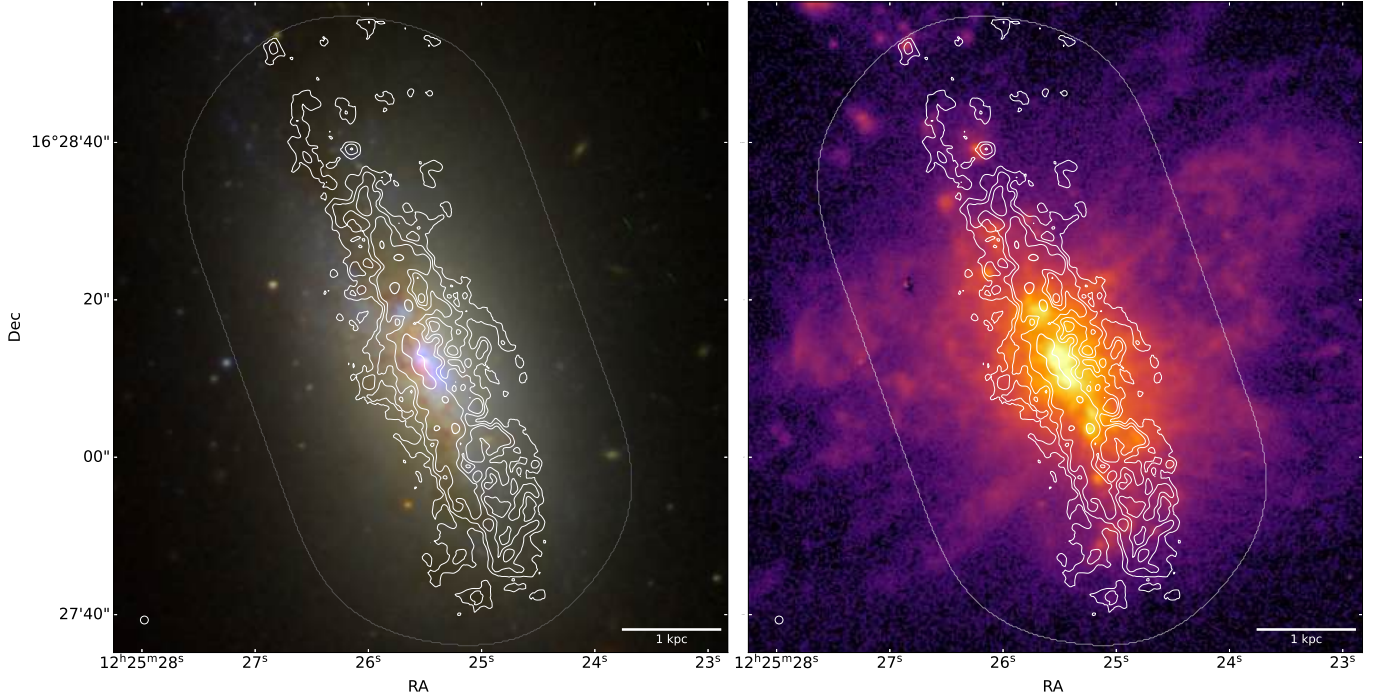


Figure 1. The CO(2-1) morphology of NGC4383. CO(2-1) intensity contours overlaid on a broad band g, i and z color composite from NGVS (left) and $H\alpha + [NII]$ net image from VESTIGE (right). Contours are shown at 2, 5, 10, 40 $K km s^{-1}$ levels. The footprint of the ALMA observations is shown in light grey.

of a rotating structure. The comparison between the two maps reveals both similarities and differences. Along the major axis, the velocity gradient of the CO appears steeper than the ionised gas. This is particularly evident on the northern side of the galaxy, where the CO disk reaches velocities lower than $1670 km s^{-1}$ ($\Delta V \sim 50 km s^{-1}$) within $\sim 600 pc$ from the centre, while the ionised phase reaches similar values only $\gtrsim 2 kpc$ from the centre. While this is at least partially due to asymmetric drift, it is also likely that the kinematics of the $H\alpha$ emission is more strongly impacted by the central starburst.

Focusing on the inner $\sim 1 kpc$ around the core of the starburst, the CO emission shows kinematics that deviates from regular rotation, most of which appear to qualitatively match features observed in the MUSE data. To better characterise the properties of these features, Figure 3 presents a series of position-velocity (PV) diagrams extracted along the major axis and five different slits parallel to the minor axis to include the most prominent features in the CO velocity field. The width of each slit is 1 arcsecond. Given the complexity of the CO velocity field, we adopt the stellar position angle of 107.5° determined by Watts *et al.* (2024) as the kinematic position angle. We tested various ranges within $\sim 10^\circ$ from this value and the results are not affected. Similarly we tested variations in the adopted kinematic centre and our results are unaffected.

Along the major axis, multiple velocity components are already evident along the line of sight, primarily within the inner 1 kpc. The presence of CO clouds kinematically detached from the disc becomes even more apparent when examining the slices taken parallel to the minor axis. Specifically, all

features marked A–F exhibit velocities that deviate from the expectations for a rotating disc, suggesting that we may be observing CO clouds entrained in the outflow and displaced from the disc. Notably, features B, C, and F appear to correspond to similar structures in the ionised-gas velocity field. These deviations, however, remain within $\sim 100 km s^{-1}$, substantially lower than the velocity range observed in the ionised phase at larger distances (Watts *et al.*, 2024).

Intriguingly, the region where the CO kinematics shows the strongest evidence for potential outflow-related anomalies matches closely with the ionisation structure of the ISM as traced by the observed ratio of the $[SII]/H\alpha$ line ratio (similar results are obtained using other line ratios). Specifically, a threshold of $\log([SII]/H\alpha) \sim -0.35$ (solid contour in the bottom panels of Fig. 2) encompasses very nicely the regions where the CO shows the strongest evidence for kinematics perturbations. This may indicate that radiation pressure from the starburst is playing a role in accelerating the clouds of CO away from the disc.

The evidence presented so far suggests that the CO disc is kinematically perturbed by the starburst occurring in the core of NGC 4383, with clear indications that some of the CO-emitting gas is being displaced up to $\sim 1 kpc$ from the disc. While it remains challenging to confidently distinguish which features are associated with the outflow and which remain part of the disc, our findings already suggest that most of the CO is unlikely to travel far from the disc. The next step is therefore to investigate whether a neutral wind exists at larger distances from the disc in the atomic phase.

As we will see, the substantially lower resolution of the H I

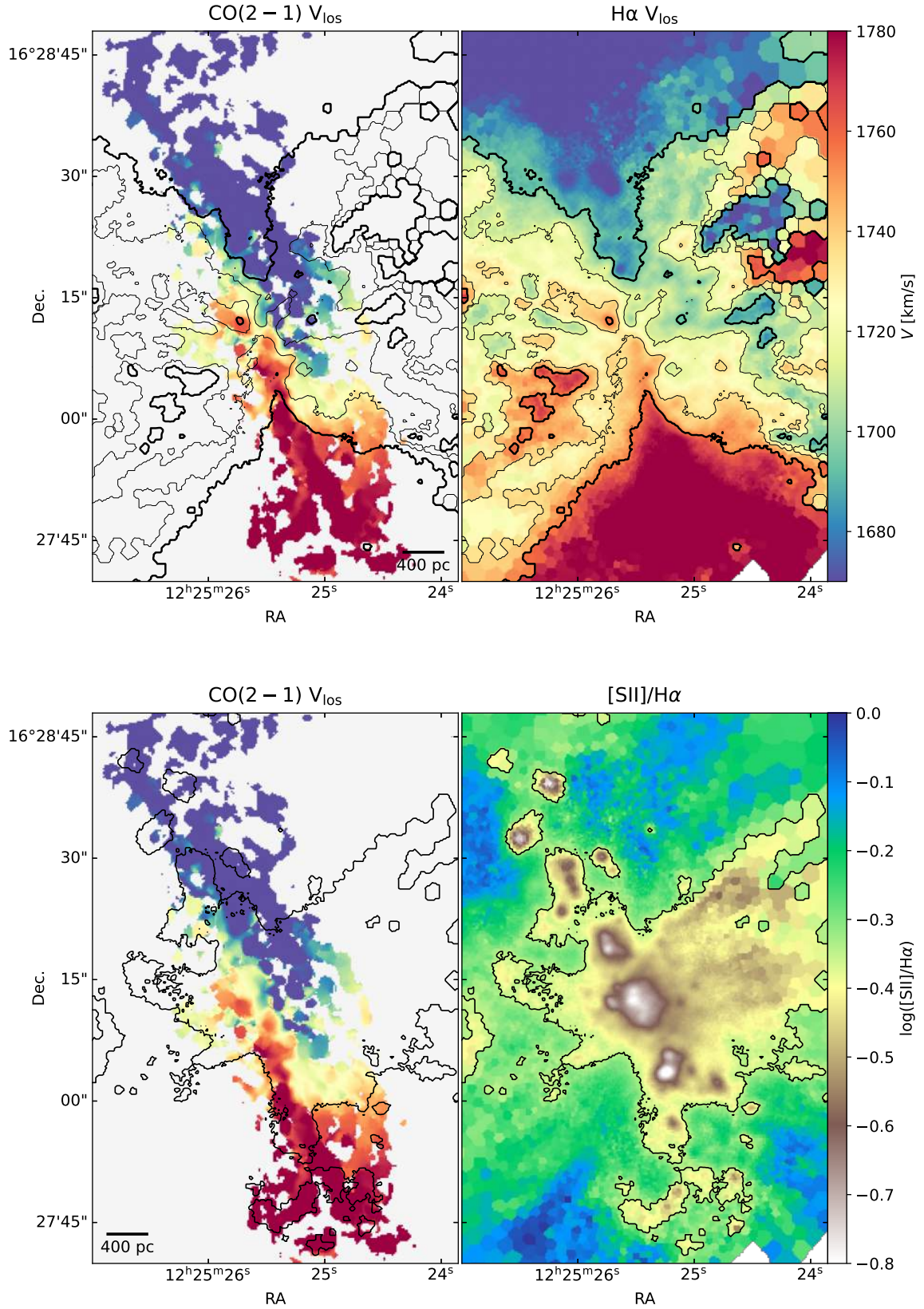


Figure 2. CO(2-1) line-of-sight velocity field. *Top:* Comparison between the CO(2-1) (left) and the MUSE H α line-of-sight velocity fields. The black contours in both maps indicate H α velocities of 1695, 1755 (tick) and 1715 and 1735 (thin) km s $^{-1}$, respectively. *Bottom:* The same CO(2-1) velocity field (left) compared to the [SII]/H α line ratio (right). The solid contour indicates $\log([SII]/H\alpha) = -0.35$.

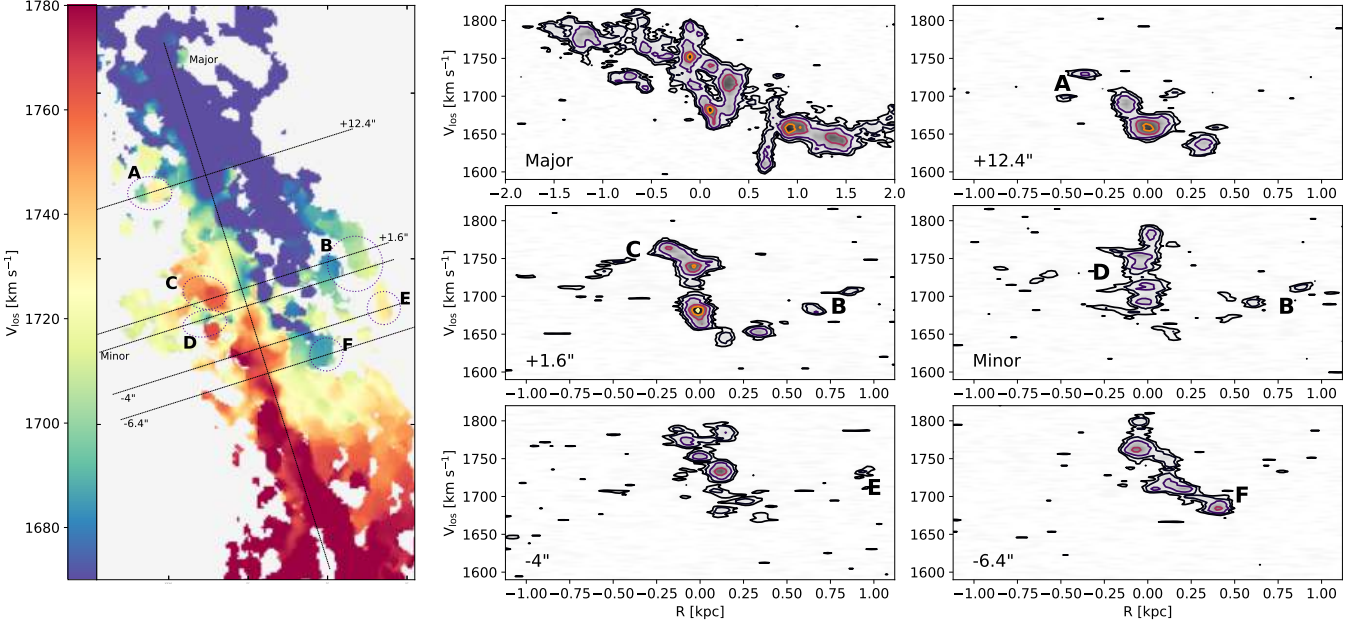


Figure 3. Position-velocity diagrams for the CO(2-1) emission. *Left:* CO(2-1) line of sight velocity field. The slits used to extract the PV diagrams are overlaid. Interesting kinematic features are indicated from A to E. *Middle and Right:* PV diagrams. The position of each slit is indicated in the bottom-left corner of each sub-plot, with regions identified in the line-of-sight velocity field highlighted with the corresponding letter.

data makes this task even more challenging. This is already apparent in Appendix 1, where we reproduce Figure 3 at a resolution of 500 pc, closer to that of the H I observations. It is clear that most, if not all, evidence of disturbed CO kinematics disappears or becomes significantly less pronounced.

3.2 Searching for the atomic phase of the outflow

The H I properties of NGC 4383 are considerably more complex than those of the molecular phase. Chung et al. (2009) previously reported the presence of an unusually extended H I disc (over 4 times larger than the optical one) when compared to the size of the optical disc, suggesting a possible accretion event as its origin. It is therefore useful to begin by zooming out and revisiting the large-scale H I distribution and kinematics of this galaxy before focusing on the region dominated by the ionised outflow. In Fig. 4, we compare the H I properties revealed by our MeerKAT data with the optical morphology (top left) and with the H α emission (bottom left). The right panels also show the H I line-of-sight velocity (V_{los} , top) and effective velocity width (σ , bottom).

The spatial distribution and kinematic properties of the H I reveal the complexity of the gas reservoir of NGC 4383. While the overall velocity field exhibits clear signs of rotation, the outer regions of the disc deviate from regular rotational motion. The most prominent feature is located at the east/north-east edge of the H I distribution, where anomalous kinematics and an increase in σ are observed. These characteristics likely result from the presence of multiple velocity components in the H I and/or an irregular warp.

All of these features were previously noted by Chung et al. (2009), and our data provide a higher-resolution view. However, the focus of this paper is on the inner, outflow-dominated

regions, which were not accessible in earlier H I interferometric studies. Within the inner ~ 5 kpc, where the H I surface density exceeds $\sim 5 \text{ M}_{\odot} \text{ pc}^{-2}$, the two-dimensional distribution of H I resembles the typical structure associated with $m = 2$ spiral patterns. GALEX ultraviolet imaging (Watts et al., 2024), as well as H α (Fig. 4, bottom-left panel), indicate recent star formation associated with these two ‘arms’. The H I surface density increases significantly toward the centre, reaching values above $30 \text{ M}_{\odot} \text{ pc}^{-2}$ in the main stellar body. This suggests that the H I disc is highly inclined, and then warps to a less inclined configuration at larger radii. As we will see, this complex kinematic structure, combined with the abundance of H I across the entire region of the ionised-gas outflow, makes it challenging to clearly identify any atomic hydrogen component unequivocally associated with the H α (and CO) outflow. In the remainder of the analysis, we will thus focus on the inner ~ 5 kpc of the H I distribution, overlapping with the bulk of the ionised-gas outflow.

Figure 5 presents a detailed comparison of the 2D intensity distribution (top), line-of-sight velocity (middle), and velocity dispersion (bottom) of the neutral atomic and ionised phases in NGC 4383. The H I exhibits two symmetric surface-density peaks along the major axis located roughly ~ 1 kpc from the centre of the outflow, with a decrease in the inner ~ 800 pc, where the brightest H α and CO emissions are observed. While this feature might be interpreted as evidence for central depletion by the outflow, the CO morphology and the fact that the ISM mass is dominated by molecular hydrogen suggest that the dip primarily reflects the predominance of the molecular phase in the central regions.

Even more intriguing is the morphology of the H I along the minor axis, which appears to form plumes that sometimes

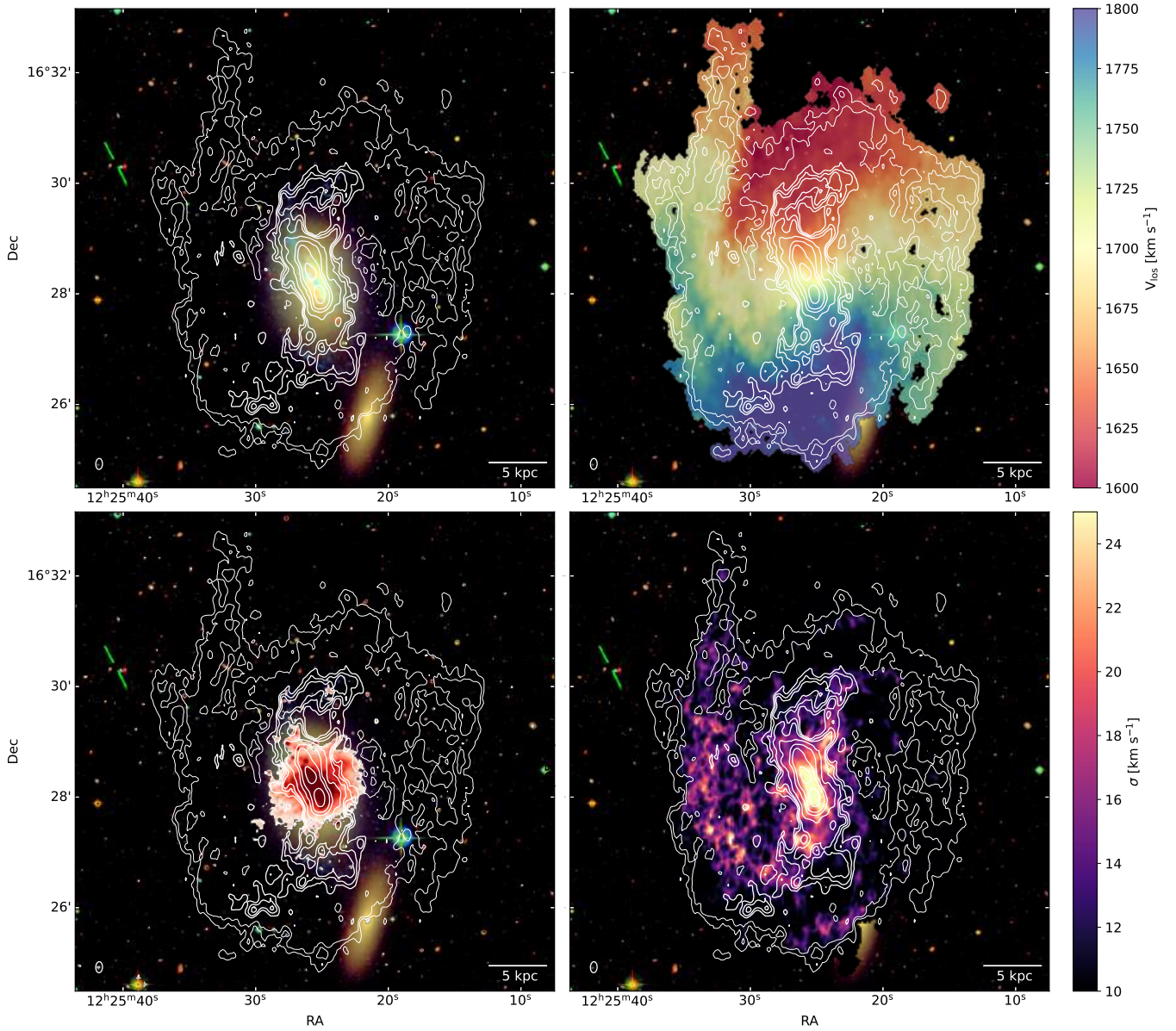


Figure 4. The H I disc of NGC4383 as revealed by MeerKAT. H I surface density contours (top-left) superposed on the H I moment 1 (top-right), VESTIGE H α + [NII] narrow-band image (bottom-left), H I effective width (bottom-right) maps. Contours are shown at 1, 2 (thin), 5, 6, 7, 10, 15, 20 and 30 (thick) $M_{\odot} \text{pc}^{-2}$. The background image shows the optical broad-band u,g,i color composite from NGVS. The synthesized beam of the H I observations is shown in the bottom-left corner of each panel.

follow the ionised-gas distribution. This is particularly notable at the northern edges of the H α outflow: to the east, the two phases align almost completely (labelled P-SE in Fig. 5 top left), whereas to the west, the H I appears offset by ~ 10 arcsec to the north relative to the ionised component (labelled P-NW in Fig. 5 top left). While this might suggest the presence of atomic hydrogen associated with the extraplanar ionised-gas outflow, Figure 5 represents only the tip of a more extended and complex H I distribution, so caution is warranted in attributing the bulk of the observed emission to outflowing gas. What is already evident, however, is that the cold ISM becomes predominantly atomic at distances of ~ 1 kpc above the main disc. As with the CO, the next step is to examine in detail the

kinematic properties of the H I and compare them with those of the ionised phase of the outflow.

In contrast to the CO, the H I line-of-sight velocity field does not exhibit the same level of kinematic disturbance observed in H α (Fig. 5, middle left), suggesting that either the H I kinematics are only weakly affected by the outflow, or, more likely, that any outflow-related effects on the moment 1 map are detectable only at spatial resolutions higher than those provided by our MeerKAT data.

The difference between the ionised and H I kinematics becomes even more evident when comparing the line broadening of the two phases. While the H α velocity dispersion (corrected for instrumental broadening) increases markedly

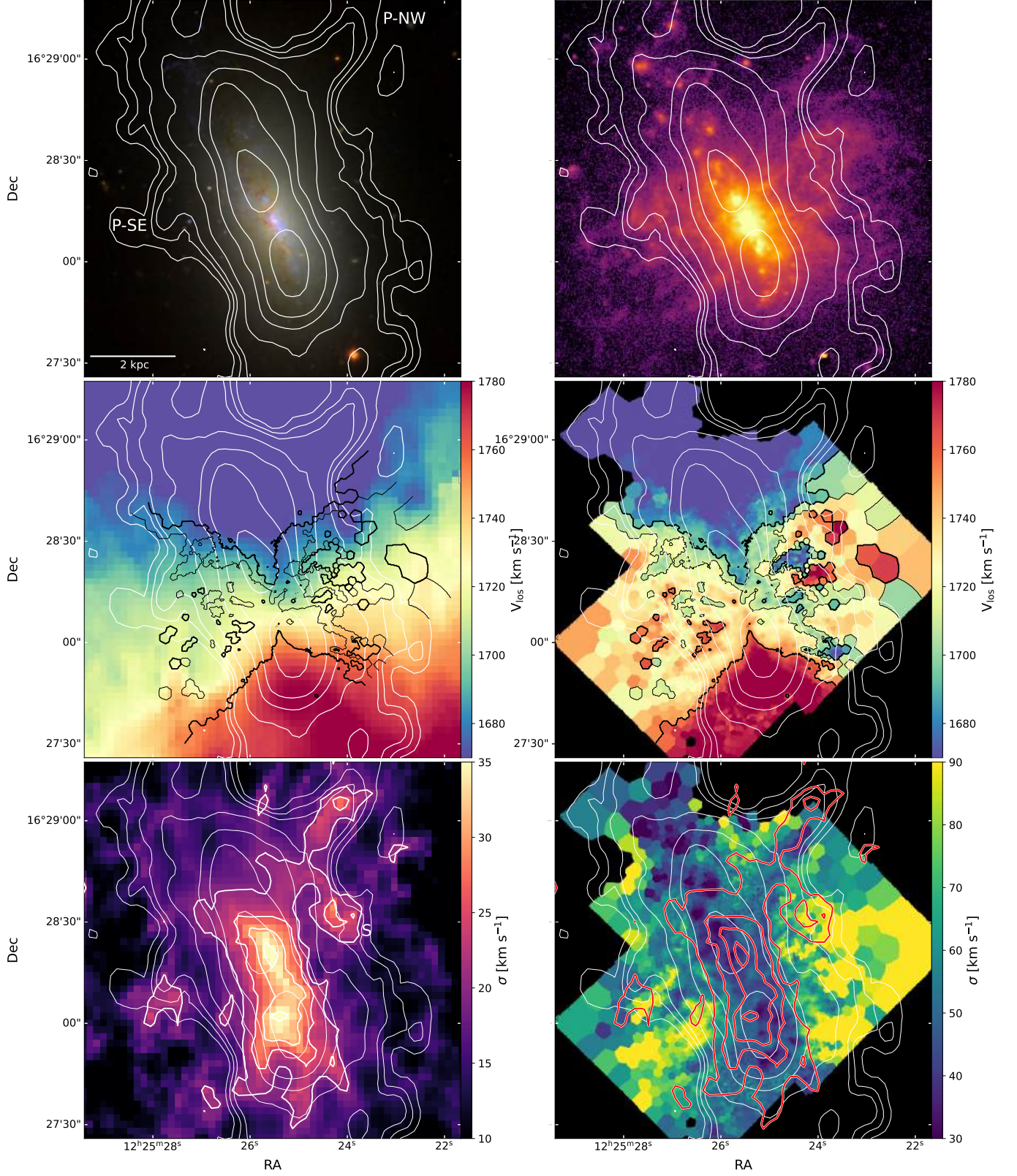


Figure 5. The H I and ionised gas outflow of NGC4383. The H I surface density contours superposed on the NGVS broad-band color image (top-left), VESTIGE H α + [NII] narrow-band image (top-right), H I moment 1 map (middle-left), MUSE H α line-of-sight velocity map (middle-right), H I effective line width (bottom-left) and H α velocity dispersion (bottom-right). The black contours in the middle column indicate H α line-of-sight velocities of 1695, 1755 (thick) and 1715 (thin) km s^{-1} . In the right column the white (top panel) and red (bottom panel) contours show H I effective width at 20, 25, 30, 35 km s^{-1} level, respectively.

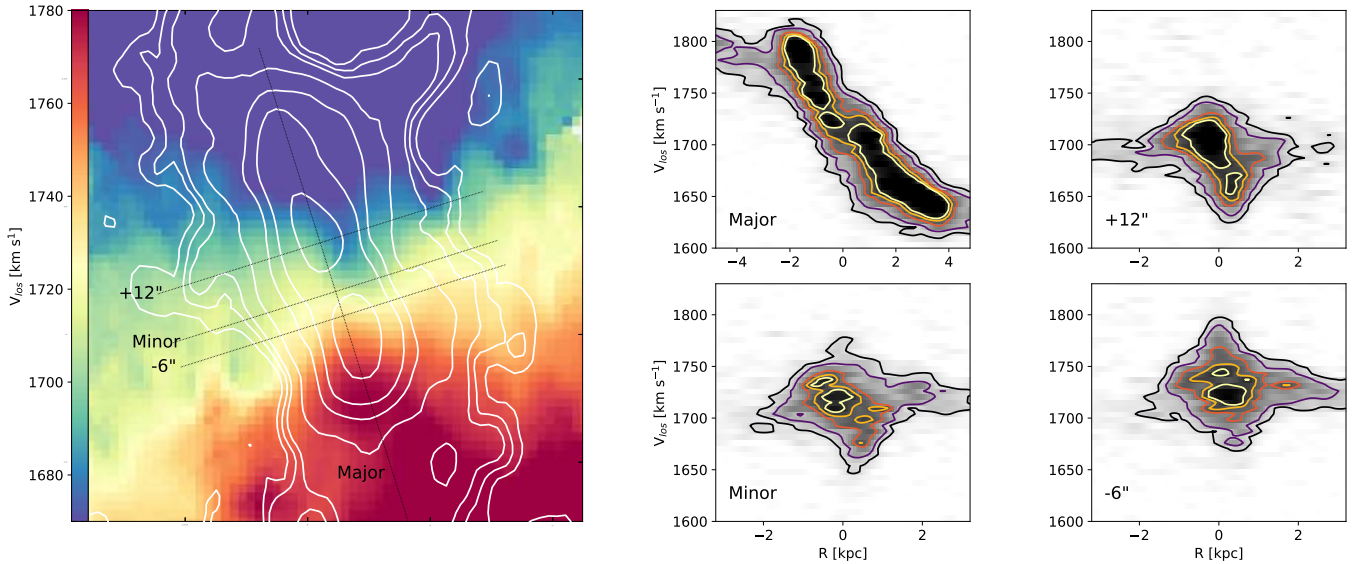


Figure 6. Position-velocity diagrams for the H I emission. *Left:* H I line of sight velocity field. The slits used to extract the PV diagrams are overlaid, with each PV diagrams shown in the middle and right column. White contours show the H I surface density distribution as in Fig. 5.

from the disc (below the MUSE spectral resolution of ~ 50 km s^{-1}) to the outflow (above 90 km s^{-1}), the H I shows the opposite trend: significant broadening (up to ~ 35 km s^{-1}) is observed within the inner ~ 1.5 kpc from the disc plane, but rapidly decreases to ~ 15 km s^{-1} , characteristic of a cold rotating disc. The regions P-NW and P-SE coincide with areas where the ionised outflow shows intermediate velocity dispersions (~ 60 km s^{-1}), with both H I plumes roughly corresponding to two of the only three locations outside the main disc where the H I line width exceeds ~ 20 km s^{-1} . Specifically, P-NW stands out clearly in the H I line-width map, whereas P-SE shows enhanced broadening only near its tip. In addition to these two plumes, we identify a third region of enhanced H I line broadening, not previously apparent in either the surface density or velocity maps. Labeled *S* in the bottom left panel of Fig. 5, this feature is projected ~ 2 kpc above the disc plane, coinciding with one of the most complex regions of ionised-gas kinematics, where line-of-sight velocities of the H α line span more than 100 km s^{-1} within a single H I beam.

To complete the characterisation of the H I kinematics of the inner regions of NGC4383, in Fig. 6 we show the position-velocity diagram for the H I along some of the same line-of-sights used for the CO. Although the lower spatial resolution of the H I data restricts us to three locations along the minor axis, we find similar deviations from ordered kinematics to what observed for the CO. At distances beyond 2 kpc, the H I shows velocities broadly consistent with disc rotation, whereas within the inner 2 kpc clear departures are observed: the southern part of the disc is redshifted and the northern part blue-shifted, consistent with the behaviour of both the H α - and CO-emitting gas. Positive velocity residuals in the SE and negative in the NW side of the outflow are also recovered if we attempt to fit the whole H I kinematics with a tilted-ring model using softwares such as 3D Barolo (Di Teodoro & Fraternali, 2015). However, the exact amount of these residuals

significantly depend on the assumptions made on the geometry of the disc and, given the complex 3D structure of the H I even in the outerparts of the gas disc, we prefer to show the data themselves without introducing any model-dependent assumption.

In summary, while the limited spatial resolution of the H I data hampers our ability to clearly separate atomic hydrogen associated with the outflow with the one belonging to the disc, we find clear evidence that part of the atomic hydrogen reservoir is directly affected by, and potentially entrained in, the ionised-gas outflow of NGC 4383 within the inner ~ 1 – 2 kpc from the plane of the disk and tantalising hints that it may extends up to at least 2.5 kpc, in particular at the edges of the cone of the ionised outflow.

4. Discussion

In this paper, we have presented new evidence that both molecular and atomic hydrogen in NGC 4383 are directly affected by the outflow, with at least part of the cold gas reservoir likely entrained in the wind. While each phase contributes differently across spatial scales, our results highlight the multi-phase nature of the outflow.

The molecular component traced by the CO(2-1) transition is detected primarily within the inner kiloparsec from the hearth of the starburst. Beyond this scale, the molecular hydrogen column density rapidly falls below our sensitivity limits, suggesting that the molecular phase contributes only marginally to the outflowing gas mass outside the central kiloparsec. Only if the α_{CO} conversion factor in the outflow would increase by >5 times the value assumed in this paper, the molecular phase could still contribute to a significant fraction of the outflow rate at larger distances. However, this seems an unlikely scenario as α_{CO} of outflowing CO-emitting gas is generally observed to be similar or lower than the Milky-Way

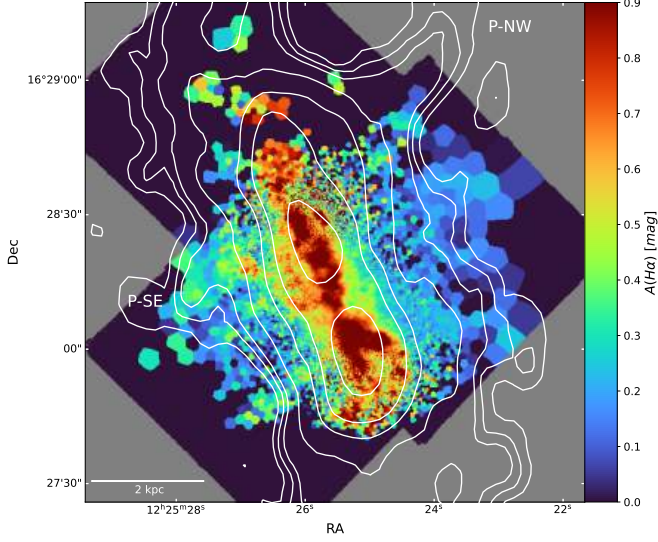


Figure 7. Evidence of dust in the outflowing gas. Map of $H\alpha$ attenuation for NGC 4383. $A(H\alpha)$ is estimated from the Balmer decrement. It is clear that the outflowing line-emitting gas is mixed with dust. White contours are from the H I intensity map as in Fig. 5.

value (e.g. Dasyra et al., 2016; Pereira-Santaella et al., 2024)

This picture seems consistent with theoretical predictions from Vijayan & Krumholz (2024), who suggest that in galaxies with average star formation rate surface densities comparable to or lower than the value observed for NGC 4383 ($\log(\Sigma_{SFR}) \sim -0.27 \text{ M}_{\odot} \text{ yr}^{-1} \text{ kpc}^{-2}$, Watts et al., 2024), the bulk of the outflowing cold material should be in the atomic phase. Nevertheless, the possibility remains that individual dense molecular clouds survive and travel to larger distances, as those observed in the Milky Way (Di Teodoro et al., 2020) and Small Magellanic Cloud (McClure-Griffiths et al., 2018), but at levels below our current sensitivity and/or resolution. Deeper millimetre observations will be required to probe this regime.

Moving to the H I, we find tantalising evidence for a cold neutral component directly connected to the outflow. We identify several extraplanar features that are both spatially and kinematically correlated with the ionised phase (and in some cases with the molecular gas as well). Despite the relatively high resolution of our MeerKAT data, the current observations still limit the extent to which these connections can be quantified, particularly given the large-scale complexity of the H I distribution in NGC 4383.

An independent way to confirm the presence of cold material in the outflow is to trace dust associated with the neutral phase. Using our MUSE data, we constructed a gas-attenuation map from the $H\alpha/H\beta$ ratio. We consider only spaxels where the $H\beta$ line has a flux greater than $3 \times 10^{-19} \text{ erg cm}^{-2} \text{ s}^{-1}$ and the ratio between line flux and the error on the line fit is larger than 5. We estimate $A(H\alpha)$ assuming Case B recombination, a temperature of 10^4 K and an electron density of 100 cm^{-2} (Osterbrock & Ferland, 2006) and a Fitzpatrick (1999) extinction curve. The gas attenuation map is shown in Fig. 7, with overlaid the H I intensity contours. We find significant extinction

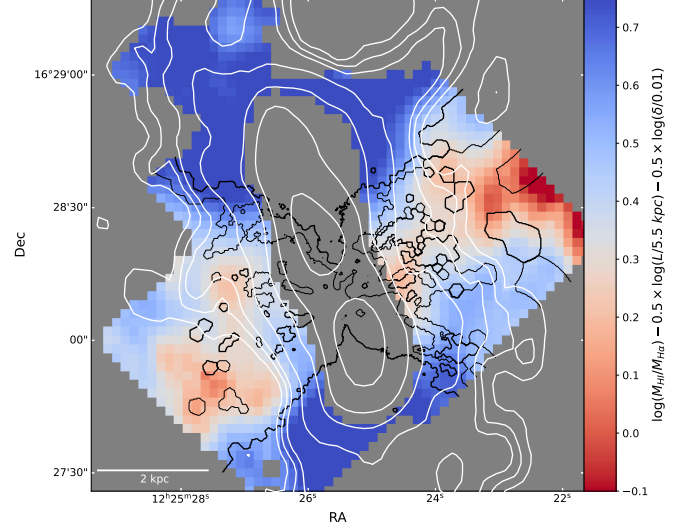


Figure 8. The mass balance between $H\alpha$ -emitting and neutral atomic gas. Map showing the mass ratio of the H I to the $H\alpha$ -emitting gas along the line of sight. The regions where the H I surface density exceeds $8 \text{ M}_{\odot} \text{ pc}^{-2}$ are masked as these are clearly dominated by H I belonging to the disc. Black contours show $H\alpha$ iso-velocity regions as in Fig. 2 to identify outflow-dominated regions. The estimate of the warm gas is highly uncertain and depends on both gas filling factor δ and the depth of the outflowing gas along the line of sight L , as indicated by the color bar. White contours are the H I intensity map as in Fig. 5.

up to projected distances of at least 2.5 kpc, with a particularly strong correspondence with the P-SE feature, and tentative associations with parts of P-NW, as well as with more diffuse structures to the southwest. This spatial match between dust attenuation and H I strongly supports the interpretation that part of the atomic hydrogen is indeed entrained in the outflow. However, the correspondence is not always perfect, suggesting that the spatial distributions of dust, H I and $H\alpha$ -emitting gas may decouple on sub-kiloparsec scales. In particular, while the bulk of the H I emission in P-SE is likely associated with outflowing material, other regions may instead be dominated by extraplanar gas in the thick disc, not directly linked to the ongoing outflow, or possibly by gas raining back toward the plane. This highlights an important caveat for direct estimates of mass-loading factors: such measurements often neglect the coexistence of gas currently entrained in the current outflow and extraplanar material projected along the same lines of sight.

Indeed, it would be tempting to use these data to present an estimate of the mass outflow rate in the different phases of NGC 4383. However, given the large uncertainty in any estimate of the amount of H I associated with the outflow, as well as in the geometry of the outflow and filling factor of the ionised gas (critical parameters for the warm gas mass outflow rate), any claim on mass outflow and loading factor could be easily off by an order of magnitude. Thus, here we prefer to use a more conservative approach and investigate what we can learn from these data despite these large uncertainties.

First, any H I component associated with the outflow shows a projected velocity lower than the $H\alpha$: where the $H\alpha$ shows line-of-sight velocities exceeding 100 km s^{-1} , the associated

H I rarely departs from systemic by more than $\sim 50 \text{ km s}^{-1}$. Of course, there could be projection effects, but it is unlikely that the geometry of the outflow in the two phases is so different to account for this. Second, the comparison of gas surface densities alone already suggests that the cold component may dominate the mass budget. Indeed, the ionised outflow mass estimated by [Watts et al. \(2024\)](#) of $M(\text{H}\alpha) = 5.1 \times 10^7 M_\odot$ over an area of 24.4 kpc^2 , implies an average ionised gas surface density of $\sim 2.1 M_\odot \text{ pc}^{-2}$, well below the typical H I surface density in the regions coincident with the outflow, which ranges between 5 and $10 M_\odot \text{ pc}^{-2}$. This would imply that the neutral gas mass is at least comparable to, and likely exceeds, the mass in the ionised phase.

However, our results have highlighted how the two phases are not always strictly co-spatial, with the H I preferentially located toward the edges of the ionised cone, consistent with scenarios in which cold clouds survive longer at the interface with the hot wind. To illustrate this, in Fig. 8 we present a map of the ratio between the mass of the H I to the one of H α -emitting gas. We reproject the H α map corrected for extinction to the pixel grid of the H I moment zero map and apply a Gaussian kernel to downgrade it to the resolution of the H I data. We then follow the same procedure described in [Watts et al. \(2024\)](#) to convert H α surface brightness into mass surface density. In addition to the gas filling factor ($\delta=0.01$), we need to make an assumption on the depth of the outflow cone along the line of sight. Here, we simply use the diameter of a cylinder with volume equal to the volume of the outflow, i.e., $L = 5.5 \text{ kpc}$. This basic assumption would primarily lead to an overestimate of the mass of gas towards the edges of the ionisation cone. Despite this, the map clearly shows that the atomic phase almost always dominates the mass budget (blue), with the H α -emitting gas reaching comparable levels only near the centre of the ionisation cone (red).

Of course, Fig. 8 should be taken with a grain of salt as the same results would be qualitatively obtained if all the H I was in a thick disc and the outflow was only in the warm phase. However, the evidence presented in this paper suggest otherwise. Module all the assumptions, it is interesting to note that the difference in mass appears larger than the difference in velocity between the two phases, suggesting that when it comes to mass loading, the atomic phase should play an important role.

Bringing these results together, the properties of NGC 4383 show striking similarities to the well-studied multiphase outflow of M82 ([Leroy et al., 2015](#)). The parallels are evident both in the relative balance between different phases, with the H I component dominating above 1–2 kpc from the disc, and in the geometry, with the cold gas preferentially located at the edges of the ionisation cone. Moreover, the velocities of the cold component are more consistent with a fountain-like flow than with a textbook large-scale galactic outflow.

Our findings add to growing evidence that in the local Universe the cold phase rarely achieves escape velocity or distances larger than a few kpc. Even in extreme cases of bipolar outflows, most of the cold gas appears to remain gravitationally bound and relatively close to the disc, ultimately falling back

and re-fueling future star formation. This is in line with the evidence that such outflows typically occur in gas-rich galaxies unlikely to exhaust their reservoirs on short timescales. NGC 4383 represents a textbook case of this flow, with the role of its multiphase outflow key for the regulating the gas cycling between the disc, halo, and intergalactic medium.

Its extended H I disc provides a substantial reservoir into which outflowing cold gas may eventually fall back, consistent with the galactic fountain scenario ([Fraternali & Binney, 2008](#)). Thus, galaxies like NGC 4383 should be viewed as extreme cases of the fountain cycle, in which gas is pushed further from the disc, making the phenomenon easier to trace observationally.

This interpretation also reinforces the likelihood that some of the extraplanar gas we detect is already in the process of returning to the disc. For example, even under a conservative assumption that the neutral gas travels at 100 km s^{-1} , it would require only $\sim 50 \text{ Myr}$ to reach 6 kpc (the full extent of the ionised outflow). Unless the starburst is both very young and short-lived, it is therefore likely that at least part of the cold material has already reached its maximum altitude and begun to fall back.

While this complicates attempts to derive reliable mass outflow rates, it also opens an exciting opportunity to directly observe cold material on its way back to the disc. Such detections have thus far eluded the community but would provide crucial constraints on the physics of gas cycling in and out of galaxies. Achieving this goal will require tracing the atomic phase at spatial resolutions below $\sim 500 \text{ pc}$, where a full three-dimensional decomposition of the H I structure could be attempted even in galaxies with complex kinematics such as NGC 4383. Excitingly, the Very Large Array is already making this possible in the Local Group ([Pingel et al., 2024](#)). While we will have to wait the phase 1 of the SKA to achieve this for large statistical samples at the distance of the Virgo cluster, projects such as MAUVE are ideally positioned to start building a statistical sample of galaxies with well-characterised multiphase outflows. This is an essential step towards clarifying the diversity of conditions that drive fountain flows.

5. Summary

In this paper, we have presented a multiphase analysis of the star-formation-driven outflow in NGC 4383, combining ALMA CO(2–1), MeerKAT H I, and VLT/MUSE data as part of the MAUVE project. Our main findings are as follows:

- We find evidence for CO clouds likely entrained in the wind close to the heart of the outflow. However, the CO emission rapidly declines with height, suggesting only a limited role of molecular gas in the mass object of the gas ejected above the disc.
- Atomic hydrogen dominates the cold component above $\sim 1 \text{ kpc}$. The H I distribution shows plumes aligned with the ionised gas. However, the extended, warped H I disc complicates unambiguous identification of outflowing gas. Independent dust extinction evidence supports the presence of cold material in the outflow.

- All observational evidence supports a scenario in which the cold phase dominates the mass budget of the material ejected a few kpc above the disc.

Overall, the multiphase properties of NGC 4383 reinforce the view that outflows in nearby gas-rich galaxies rarely expel cold gas permanently, but instead cycle material between disc and halo. This highlights the importance of tracing all gas phases at high spatial resolution to fully characterise the baryon cycle in galaxies.

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In addition to the softwares listed in the text, this study has made extensive use of PYTHON (Van Rossum & Drake, 2009) libraries — ASTROPY (Astropy Collaboration et al., 2013, 2018, 2022), NUMPY (Harris et al., 2020), SCIPY (Virtanen et al., 2020), MATPLOTLIB (Hunter, 2007), JUPYTER (Kluyver et al., 2016) and EXTINCTION.

Funding Statement

Data Availability Statement

Author Contributions

Appendix 1. CO(2-1) position-velocity diagram at 500 par-sec resolution.

To illustrate the dramatic impact of spatial resolution on our ability to identify kinematic disturbances, in Fig. 9 we reproduce Fig. 3 for a cube degraded to a resolution of 500 pc. Although this resolution remains higher than that of the MeerKAT data, it is evident that almost all signatures of disturbed kinematics disappear.

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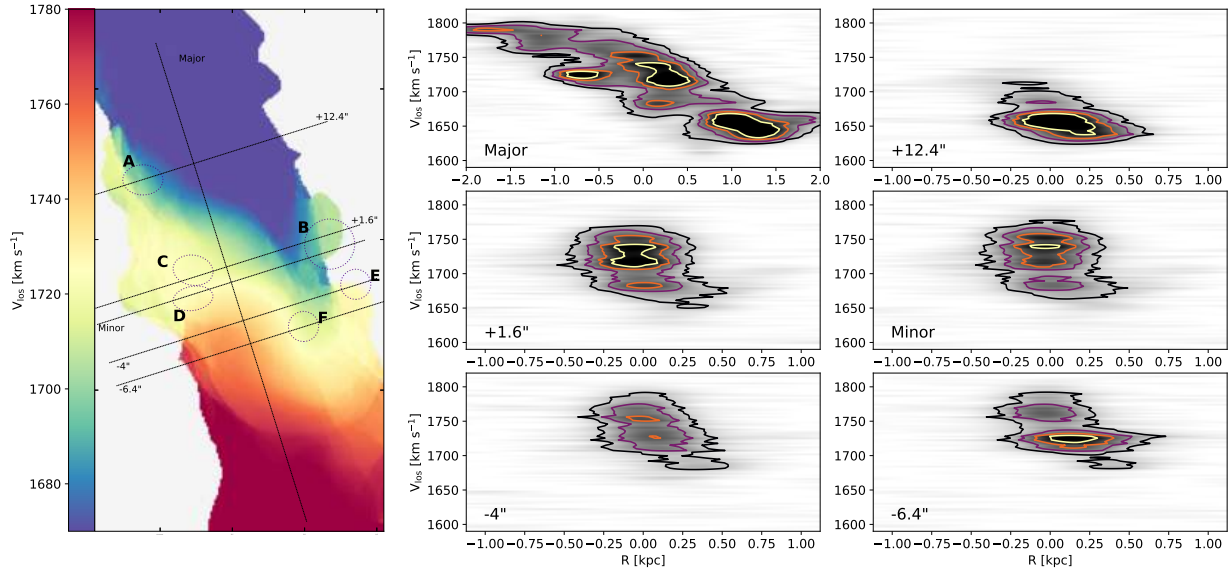


Figure 9. Position-velocity diagrams for the CO(2-1) emission at 500 parsec resolution. Same as Fig.3 but for a cube degraded to a 500 parsec resolution.

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